Stability thresholds of N-vortex rings on the hyperbolic plane (tanh model): rational Fourier blocks, palindromic threshold polynomials, and a certified table for $N \le 16$

Gordon Speagle
Independent Researcher, Boise, ID, USA

DRAFT — AI-reviewed only; no human specialist review

2026-10-11 (revised)

Abstract

Montaldi and Tokieda [1, §4.2] considered the image Green's function on the hyperbolic plane, whose pair interaction is $-\ln\tanh(d/2)$ (the Hamiltonian they attribute to Kimura), and wrote that "the stability problem seems no longer tractable analytically". Boatto [2] had earlier computed the stability ranges of the regular N-ring in this model numerically. We show that the linear problem is tractable. In log-polar coordinates the Hessian of the augmented Hamiltonian at the ring splits into 2×2 Hermitian Fourier blocks whose entries are explicit rational functions of $\xi = \tanh^2(r/2)$, where r is the hyperbolic ring radius. Unlike the "sinh" model, the blocks have a radial–tangential coupling. Under $\xi \mapsto 1/\xi$ the diagonal entries are invariant and the coupling changes sign; this reciprocity comes from inversion in the boundary circle. It makes the threshold polynomials palindromic. For $N = 8, \dots, 16$ we certify the spectral threshold $\xi_*(N)$ (exact rational root isolation and Arb ball arithmetic) and the critical mode $\lfloor N/2 \rfloor$. We also certify, with PARI/GP and an independent mod-p factorisation test, that $\xi_*(N)$ has an irreducible palindromic minimal polynomial of degree $2N-2$, $\frac{3}{2}N-2$ or $\frac{3}{2}N-3$ according to N odd, $N \equiv 0$ or $N \equiv 2 \pmod 4$. The number $\xi_*(8)$ is an algebraic unit of degree 10; for $9 \le N \le 16$, $\xi_*(N)$ is not an algebraic integer. (That the thresholds are algebraic is automatic once the blocks are rational; the content is their explicit structure.) Rings with $N \le 7$ are stable at every radius. For odd N the energy–momentum (Hessian-definiteness) threshold lies strictly above the spectral threshold. In the gap the ring is linearly (spectrally) stable, and its nonlinear stability is undecided (a Krein window). Numerically $\cosh r_*/N$ approaches $\kappa_\infty \approx 0.1343122$, the root of a transcendental equation obtained by a formal limit; the convergence is not proved. In the sinh model, $\cosh r_* \sim \sqrt{N}/2$.

1 Motivation and prior work

The question left open by Montaldi–Tokieda. In §4.2 of [1] (arXiv:1210.5662v1, p. 25), for the Green's function $G = \log \frac{|z-w|^2}{|1+\lambda z\bar{w}|^2}$, Montaldi and Tokieda write: "On the hyperbolic plane $\lambda < 0$ this reduces to the Hamiltonian adopted by Kimura [10]." After discussing $\lambda > 0$, they write:

> "The calculations based on this option of Green's function get so cumbersome that the stability problem seems no longer tractable analytically. It seems likely that the results are similar to those in section 3, though the details of where the bifurcations occur will differ."

At $\lambda = -1$ this Green's function is $\log \tanh^2(d/2)$, i.e. the model (1) below. Our results answer the first sentence, for linear stability. The Hessian at the ring reduces to explicit rational 2×2 blocks (Thm. 2). The threshold is the root of an explicit polynomial equation (§6), and for $N \leq 16$ it is certified together with its minimal polynomial (§7). On the guess in the second sentence: the stability picture of their §2 carries over qualitatively. Rings with $N \leq 7$ are stable at every radius; for $N \geq 8$ a ring is stable exactly beyond a threshold radius, and the critical mode is $\lfloor N/2 \rfloor$. There is one structural difference. The tanh blocks have a radial–tangential coupling, and for odd N it separates the spectral threshold from the energy–momentum threshold (§8); this has no counterpart in the sinh model. We do not study the bifurcations themselves, which are the subject of their §3. So their guess about the bifurcation types remains open.

Prior work on this model. The record of what we read is in `literature.md`.

- Boatto [2] studies exactly this model: on the hyperbolic surface her Green's function is "$\Psi_K = -\frac{1}{2\pi} \ln \tanh(\frac{\sqrt{|K|}}{2} r)$". She writes the Hessian in the rotating frame in block form with "$\mathcal{Q}$ and $\mathcal{P}$ ... symmetric circulant matrices, while $\mathcal{G}$ is an asymmetric circulant matrix", i.e. the Fourier-block structure with a coupling term. She then "(numerically) monitor[s] the maximum and the minimum eigenvalues" and concludes that "for $N \geq 7$, we can only assure stability for sufficiently negative K" (her Fig. 5a, $N = 5, \ldots, 9$, ring of geodesic radius 1). So the stability question, the block structure and the qualitative picture for $N \geq 8$ are hers. We did not find in her paper closed forms for the block entries, threshold values beyond plotting accuracy, the algebraic structure of the thresholds, or the odd-N separation of the two thresholds. Her criterion is Hessian definiteness, i.e. our energy–momentum threshold. Her ring has fixed geodesic radius 1 while K varies, so $K = -r^2$ with $\xi = \tanh^2(r/2)$. Implemented as written, her equations (14), (17) and (31) reproduce our thresholds exactly: $K_*(8) = -0.28218454857650$, i.e. $\xi_*(8)$ to 10^{-41}, and $K_{\mathrm{EM}}(9) = -0.48869837$, i.e. our $\xi_{\mathrm{EM}}(9)$, including the certified gap $\xi_{\mathrm{EM}} - \xi_* = 3.22 \cdot 10^{-8}$. Her Fourier blocks equal ours up to m-independent factors. The zero crossings in her Fig. 5a lie slightly elsewhere, at $K \approx -0.2465$ ($N = 8$) and -0.3964 ($N = 9$), digitised to ± 0.002. These are reproduced to plotting accuracy by the pair law $-\ln \sinh D$ (D the full geodesic distance), i.e. her eq. (17) with the sign of the $\ln(d+1)$ term flipped, which gives -0.24656 and -0.39692. The difference therefore seems to lie in the computation behind the plotted $K < 0$ curves rather than in the model, and it leaves her qualitative conclusions unchanged; her $K > 0$ curves agree with her equations. We may of course have misread her conventions. Details and scripts: `tanh/boatto_check/`.

- Montaldi–Nava-Gaxiola [3] and Nava-Gaxiola's thesis [4] use the same Hamiltonian but analyse only $N = 2, 3$. Both point to Hwang–Kim [6] for rings.

- Hwang–Kim 2013 [6]: we saw only the abstract ("the configuration of rings of vortices is investigated in detail"). According to [4] it uses the same Hamiltonian and gives conditions for ring relative equilibria. Whether it contains stability thresholds is unknown.

- Kimura [5]: abstract only.

Hence the closed-form blocks, the reciprocity, the certified table, the minimal polynomials and the odd-N window are, as far as we can tell, new, subject to Hwang–Kim 2013, which we have not read.

2 Scope and status labels

The companion note `sinh/note.tex` treats the sinh model, where everything is known [1]. We use these labels:

PROVED (all N)	Fourier-block theorem (Thm. 2); linearisation and stability criteria (Prop. 3); reciprocity (Prop. 5); threshold equation and even-N form.
CERTIFIED (exact arithmetic, $N \le 16$)	$b_m > 0$ on $(0,1)$; critical mode $\lfloor N/2 \rfloor$; the value of $\xi_*(N)$ and the sign change of a_{crit} at ξ_*; $N \le 7$ stable for all r; odd-N energy–momentum threshold strictly above ξ_*. Exact rational root isolation (sympy) and Arb balls (python-flint), §7.
CERTIFIED by computer algebra (PARI/GP), $8 \le N \le 16$	Irreducibility over $\mathbb{Q}$ of the threshold polynomial f_N, hence: f_N is the minimal polynomial of $\xi_*(N)$, the degrees (4) and leading coefficients of Table 1, palindromy of the minimal polynomial, $\xi_*(8)$ a unit, $\xi_*(N)$ not an algebraic integer for $N \ge 9$ (§7.1). Not checked in Lean.
Lean (checked in the repository's CI build)	In `lean/Havelock/TanhAlgebra.lean`. Taking the block formulas of Thm. 2 as definitions, Lean checks: (i) reciprocity of a_m and the even-N threshold equation; (ii) for every $2 \le N \le 16$, the statements of Thm. 7 on positivity, sign change and critical mode, with ξ_* located to within 10^{-20} in a rational interval; (iii) $\xi_*(8)$ is an algebraic integer and a unit; (iv) the odd-N Krein window is non-empty for $N = 9, 11, 13, 15$. Lean does *not* check irreducibility (that is the PARI item above), the derivation of the blocks, or the dynamics.
CONJECTURE	the degree pattern (4) and irreducibility/palindromy for all N; $b_m > 0$ and critical mode $\lfloor N/2 \rfloor$ for all N.
NUMERICAL	κ_∞: a numerically evaluated root of an explicit equation obtained by a formal limit; its identification with $\lim \cosh r_*/N$ is unproved. The $1/N$ correction in §9 is a fit. Critical mode up to $N = 800$.

Nonlinear (Lyapunov) stability is claimed only in the form "positive-definite Hessian on the non-rotational modes". The passage from this to Lyapunov stability holds *modulo* $SO(2,1)$; see Remark 4. We have not checked the hypotheses of an energy–momentum theorem for this model against a source.

3 The model

3.1 Hamiltonian

We use the Poincaré disk $\mathbb{D} = \{|z| < 1\}$ with the metric $4|dz|^2/(1-|z|^2)^2$ and the area (symplectic) form $\omega = \sum_k \lambda(z_k)\, dx_k \wedge dy_k$, $\lambda(z) = 4/(1-|z|^2)^2$. The geodesic distance satisfies $\tanh(d(z,w)/2) =$

$|z-w|/|1-z\bar{w}|$. For N unit vortices, take

$$H = -\sum_{j<k} \ln\tanh\frac{d_{jk}}{2} = \sum_{j<k}\Big(-\ln|z_j-z_k|+\ln|1-z_j\bar{z}_k|\Big), \qquad J = \sum_k \frac{1+|z_k|^2}{1-|z_k|^2} = \sum_k \cosh r_k. \quad (1)$$

Here J generates rotations about 0, and Hamilton's equations are $\dot{x}_k = \lambda^{-1}\partial_{y_k}H$, $\dot{y}_k = -\lambda^{-1}\partial_{x_k}H$, as in `note.tex`. The function $f(d) = \ln\tanh(d/2)$ satisfies $f'' + \coth d\, f' = 0$ for $d > 0$ (since $f' = 1/\sinh d$). So $G(d) = -\frac{1}{2\pi}\ln\tanh(d/2)$ is the Green's function of $\Delta_{\mathbb{H}^2}$ that vanishes as $d \to \infty$.

Attribution (secondary only). Montaldi–Tokieda [1, §4.2] state that the image Green's function $\log|z-w|^2/|1+\lambda z\bar{w}|^2$ "on the hyperbolic plane $\lambda < 0$ ... reduces to the Hamiltonian adopted by Kimura". Montaldi–Nava-Gaxiola [3, eq. (12)] write Kimura's Hamiltonian as $-\frac{1}{4\pi}\sum \Gamma_i\Gamma_j \ln\frac{\langle X_i,X_j\rangle+1}{\langle X_i,X_j\rangle-1}$ with $\langle X,Y\rangle = -\cosh d$, which equals $-\frac{1}{4\pi}\sum \ln\tanh^2(d/2)$. Boatto [2] uses the same Green's function on the hyperbolic surface. We have read only the abstract of Kimura [5]. Hence (1) is "the model attributed to Kimura", and we quote nothing from Kimura directly.

No Robin term (intrinsic remark). $\mathbb{H}^2$ is homogeneous, so the regular part of G at the diagonal, measured in the intrinsic metric, is the constant $\lim_{d\to 0}(G(d) + \frac{1}{2\pi}\ln d) = \frac{1}{2\pi}\ln 2$. The Robin (self-interaction) function is therefore constant, and (1) has no one-body self-term. This contrasts with the flat disk with a hard wall (§10), where the Robin function $\frac{1}{2\pi}\ln(1-|z|^2)$ is not constant and, combined with the Euclidean area form, changes the dynamics. In conformal coordinates, a one-body term $-\frac{1}{2}\sum\ln(1-|z_k|^2)$ does appear in (2) below. That term comes from the $j \neq k$ image sums, not from a self-interaction.

Relation to the sinh model. Since $\sinh(d/2) = \tanh(d/2)\cosh(d/2)$, $H_{\sinh} = -\sum\ln\sinh(d_{jk}/2) = H - \sum_{j<k}\ln\cosh(d_{jk}/2)$. A direct computation gives $\Delta_{\mathbb{H}^2}\ln\cosh(d/2) = \frac{1}{2}$. So the sinh model is the tanh model plus a uniform compensating background vorticity attached to each vortex. This is the continuation to $\mathbb{H}^2$ of the sphere model, where Gauss's law forces such a term. The sinh model is the $\lambda < 0$ member of Montaldi–Tokieda's family [1, (1.12)].

3.2 Power-sum decomposition

Let $P_p = \sum_j z_j^p$. Using $\ln|1-u| = -\operatorname{Re}\sum_{p\geq 1} u^p/p$ and $\sum_{j\neq k}(z_j\bar{z}_k)^p = |P_p|^2 - \sum_j |z_j|^{2p}$, we get

$$H = -\sum_{j<k}\ln|z_j-z_k| \;-\; \frac{1}{2}\sum_j \ln(1-|z_j|^2) \;-\; \frac{1}{2}\sum_{p\geq 1}\frac{|P_p|^2}{p}. \quad (2)$$

The series converges absolutely on $\mathbb{D}^N$. The last term $\Phi := -\frac{1}{2}\sum_p |P_p|^2/p$ is the source of the apparent intractability. On a ring it is diagonal in the Fourier modes.

4 Fourier blocks

4.1 Coordinates and the relative equilibrium

Let $w = e^{2\pi i/N}$ and write $z_j = \sqrt{\xi}\, w^j e^{s_j + i\theta_j}$, so the ring of Euclidean radius $\sqrt{\xi}$ is $s = \theta = 0$. Then $\omega = \sum_j \kappa(s_j)\, ds_j \wedge d\theta_j$ with $\kappa(s) = 4\xi e^{2s}/(1-\xi e^{2s})^2$, and $\kappa_0 := \kappa(0) = 4\xi/(1-\xi)^2$. Set

$$c := \cosh r = \frac{1+\xi}{1-\xi}, \qquad X := \xi^N, \qquad W_k := \sum_{\substack{p\geq 1 \\ p\equiv k \ (N)}} p\,\xi^p = \frac{k\xi^k}{1-X} + \frac{N\xi^{k+N}}{(1-X)^2} \quad (1 \leq k \leq N).$$

In particular $W_N = NX/(1-X)^2$. Let $F = H - \Omega J$.

Lemma 1. *The ring is a relative equilibrium for every $\xi \in (0,1)$, with $\Omega\kappa_0 = -\dfrac{N-1}{2} + \dfrac{\xi}{1-\xi} - \dfrac{NX}{1-X}$. At the ring, $dF = 0$ in all $2N$ variables.*

Proof. All θ-derivatives vanish by the dihedral symmetry. For s_j: the pair term contributes $-\frac{N-1}{2}$ (its linear part is $-\frac{1}{2}\sum_{j<k}(s_j + s_k)$; see `note.tex`, §2). The one-body part is $V(s) = -\frac{1}{2}\ln(1-u) - \Omega\frac{1+u}{1-u}$ with $u = \xi e^{2s}$, and $V'(0) = \frac{\xi}{1-\xi} - \Omega\kappa_0$. For Φ, write $|P_p|^2 = \xi^p \sum_{j,k} w^{(j-k)p} e^{p\epsilon_j + p\bar{\epsilon}_k}$ with $\epsilon_j = s_j + i\theta_j$. Then $\partial_{s_j}|P_p|^2\big|_0 = 2p\,\xi^p \operatorname{Re}\sum_k w^{(j-k)p} = 2pN\xi^p[N \mid p]$, so $\partial_{s_j}\Phi|_0 = -N\sum_{q\geq 1} X^q = -NX/(1-X)$. Setting the sum to zero gives Ω. □

4.2 The block theorem

Theorem 2 (Fourier blocks; PROVED)**.** *In the coordinates (s,θ), the Hessian of F at the ring is block-circulant. On the Fourier mode $(s_j,\theta_j) = (S,T)\,w^{mj}$, $1 \leq m \leq N-1$, it acts as the Hermitian matrix*

$$M_m = \begin{pmatrix} a_m & ig_m \\ -ig_m & b_m \end{pmatrix},$$

where

$$\begin{aligned} a_m &= (N-1)c - \tfrac{1}{2}(c-1)^2 - \tfrac{m(N-m)}{2} + \frac{2NcX}{1-X} - \frac{N^2X}{(1-X)^2} \\ &\quad - \frac{N}{2}\,\frac{m\xi^m + (N-m)\xi^{N-m} + (N-m)\xi^{N+m} + m\xi^{2N-m}}{(1-X)^2}, \\ b_m &= \tfrac{m(N-m)}{2} + NW_N - \tfrac{N}{2}\,(W_m + W_{N-m}), \qquad g_m = \tfrac{N}{2}\,(W_m - W_{N-m}). \end{aligned}$$

On mode 0, $b_0 = g_0 = 0$. All entries are rational functions of ξ with rational coefficients. Moreover $a_{N-m} = a_m$, $b_{N-m} = b_m$, $g_{N-m} = -g_m$, and $g_{N/2} = 0$ for even N.

For comparison, the sinh model has $a_m = (N-1)(1+c^2)/2 - m(N-m)/2$, $b_m = m(N-m)/2$ and $g_m = 0$ (`note.tex`, Prop. 2).

Proof. Since $dF = 0$ at the ring (Lemma 1), the Hessian is well defined. We add the three contributions in (2) and the term $-\Omega J$.

Pair term. As in `note.tex` (Lemma 1 and the expansion before it), $-\sum_{j<k}\ln|z_j - z_k|$ has Hessian $-\frac{m(N-m)}{2}$ in ss and $+\frac{m(N-m)}{2}$ in $\theta\theta$ on mode m, with no $s\theta$ coupling.

One-body terms. $V(s)$ from the proof of Lemma 1 does not depend on θ. Using $\frac{d}{ds} = 2u\frac{d}{du}$,

$$V''(0) = \frac{2\xi}{(1-\xi)^2} - \frac{8\Omega\xi(1+\xi)}{(1-\xi)^3} = \frac{2\xi}{(1-\xi)^2} - 2c\,\Omega\kappa_0.$$

Substituting $\Omega\kappa_0$ and using $c - 1 = 2\xi/(1-\xi)$, so that $\frac{2\xi}{(1-\xi)^2} - \frac{2c\xi}{1-\xi} = -\frac{1}{2}(c-1)^2$, gives $V''(0) = (N-1)c - \frac{1}{2}(c-1)^2 + \frac{2NcX}{1-X}$. This term is the same on every mode and lies in the ss entry.

Power sums. The second-order part of $e^{p\epsilon_j + p\bar{\epsilon}_k}$ is $\frac{p^2}{2}(\epsilon_j^2 + 2\epsilon_j\bar{\epsilon}_k + \bar{\epsilon}_k^2)$. Summing with the weight $w^{(j-k)p}$: (i) the squares give $p^2 N[N \mid p]\sum_j \mathrm{Re}\,\epsilon_j^2 = p^2 N[N \mid p]\sum_j (s_j^2 - \theta_j^2)$; (ii) the cross term gives $p^2\big|\sum_j w^{jp}\epsilon_j\big|^2$. Hence the quadratic part of Φ is

$$\Phi_2 = -\frac{N}{2}W_N\sum_j(s_j^2 - \theta_j^2) \;-\; \frac{1}{2}\sum_{p\geq 1} p\,\xi^p\Big|\sum_j w^{jp}(s_j + i\theta_j)\Big|^2.$$

The first piece contributes $-NW_N$ to ss and $+NW_N$ to $\theta\theta$. For the second, expand $|\sum_j w^{jp}\epsilon_j|^2$ as a real quadratic form. It equals $\sum_{j,k}\cos\phi_{jk}(s_j s_k + \theta_j\theta_k) + 2\sum_{j,k}\sin\phi_{jk}\, s_j\theta_k$ with $\phi_{jk} = 2\pi(j-k)p/N$. Its Hessian has circulant blocks $2\cos\phi_{jk}$ $(ss,\,\theta\theta)$ and $2\sin\phi_{jk}$ $(s\theta)$. On the vector w^{mk} their symbols are $N([p \equiv m] + [p \equiv -m])$ and $-iN([p \equiv m] - [p \equiv -m])$, using $\sum_k w^{(j-k)p}w^{mk} = Nw^{mj}[p \equiv m]$. Multiplying by $-\frac{1}{2}p\xi^p$ and summing over p gives $-\frac{N}{2}(W_m + W_{N-m})$ on both diagonal entries and $+\frac{iN}{2}(W_m - W_{N-m}) = ig_m$ in the $s\theta$ entry.

Adding the three pieces gives a_m, b_m and g_m. In particular, $a_m = V''(0) - \frac{m(N-m)}{2} - NW_N - \frac{N}{2}(W_m + W_{N-m})$, and expanding $W_m + W_{N-m}$ over the common denominator $(1-X)^2$ gives the bracket in the statement. For $m = 0$, W_0 should be read as W_N, which gives $b_0 = NW_N - NW_N = 0$. The symmetries follow from $m \leftrightarrow N - m$. □

Checks. The blocks equal the Fourier symbol of the exact Cartesian Hessian of F (sympy second derivatives, with Ω solved numerically) to $\leq 2\cdot 10^{-48}$. The grid was $N \in \{3,5,7,8,9,10,12,13,16\}$ and $\xi \in \{0.02, 0.09, 0.3, 0.75\}$ (`check_jacobian.out`). An independent second reader built a Cartesian Hessian at 60–110 digits without seeing these formulas and matched the blocks to $2.5\cdot 10^{-59}$ (`review/`). The compact form of a_m agrees exactly with the block formula for $N = 3..16$ and all m (`identities.out`).

5 Linearisation and stability criteria

Proposition 3 (PROVED)**.** *On mode m the linearised flow is $\kappa_0(\dot{S},\dot{T})^\top = \left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)M_m(S,T)^\top$. Its eigenvalues are*

$$\kappa_0\sigma = -ig_m \pm \sqrt{-a_m b_m}.$$

Hence the ring is spectrally stable iff $a_m b_m \geq 0$ for all $1 \leq m \leq N-1$. The Hessian of F is positive definite on the non-rotational modes $1 \leq m \leq N-1$ iff $a_m > 0$ and $a_m b_m - g_m^2 > 0$ for all such m.

Proof. $\det\left(\left(\begin{smallmatrix}0 & 1\\ -1 & 0\end{smallmatrix}\right)M_m - \lambda\right) = (\lambda + ig_m)^2 + a_m b_m$. A 2×2 Hermitian matrix is positive definite iff its $(1,1)$ entry and its determinant $a_m b_m - g_m^2$ are positive. Mode 0 (rotation, and change of ξ along the family) carries $b_0 = 0$ and is excluded. □

So the coupling g_m only shifts the frequencies; it does not affect spectral stability. It does affect definiteness, and this is the origin of the odd-N window in §8.

Remark 4 (energy–momentum, "modulo $SO(2,1)$"). The variation dJ annihilates every mode $m \neq 0$. Hence $\ker dJ \subset (\text{modes } 1..N-1) \oplus \mathbb{R}\,\partial_\theta^{(0)}$, and positive definiteness on modes $1..N-1$ implies definiteness of d^2F on $\ker dJ$ modulo the rotation orbit. The full symmetry group of (1) is the isometry group $PSL(2,\mathbb{R}) \cong SO(2,1)_0$. Its momentum map has three components, and the modes $m = \pm 1$ are tangent to the non-compact (boost) directions. We therefore state any Lyapunov conclusion only *modulo* $SO(2,1)$: up to motion along the group orbit. We do not claim stability modulo $SO(2)$ alone. We did not verify the hypotheses of a specific energy–momentum theorem (for example Montaldi–Tokieda [1, Thm. 2], which is stated for the sinh model) for the tanh model.

6 Reciprocity and the threshold equation

Proposition 5 (reciprocity; PROVED for all N, m)**.** *As rational functions of ξ, under the map $\xi \mapsto 1/\xi$,*

$$a_m(1/\xi) = a_m(\xi), \qquad b_m(1/\xi) = b_m(\xi), \qquad g_m(1/\xi) = -g_m(\xi).$$

Hence $M_m(1/\xi) = \overline{M_m(\xi)}$, and $a_m b_m$ and $a_m b_m - g_m^2$ are invariant.

Proof. For $1 \le k \le N-1$ the closed form of W_k gives $W_k(1/\xi) = \xi^{N-k}\frac{(N-k)+kX}{(1-X)^2} = W_{N-k}(\xi)$, and $W_N(1/\xi) = W_N(\xi)$ since $X/(1-X)^2$ is invariant under $X \mapsto 1/X$. So $W_m + W_{N-m}$ is invariant and $W_m - W_{N-m}$ changes sign, which gives the claims for b_m and g_m. For a_m: under $\xi \mapsto 1/\xi$ we have $c \mapsto -c$ and $X \mapsto 1/X$. The one-body part changes as follows: $(N-1)c \mapsto -(N-1)c$; $-\frac{1}{2}(c-1)^2 \mapsto -\frac{1}{2}(c+1)^2$, a change of $-2c$; and $\frac{2NcX}{1-X} \mapsto \frac{-2Nc/X}{1-1/X} = \frac{2Nc}{1-X}$, a change of $+2Nc$. The total change is $-2(N-1)c - 2c + 2Nc = 0$. The term $N^2X/(1-X)^2$ is invariant. In the last fraction, the numerator is multiplied by ξ^{-2N} (its exponents $m, N-m, N+m, 2N-m$ are permuted by $e \mapsto 2N-e$) and so is $(1-X)^2$. The term $m(N-m)/2$ is constant. □

A symbolic check treating ξ, $X = \xi^N$ and $t = \xi^m$ as independent symbols confirms the identity for a_m (`review/palin_symbolic.py`); Lean checks it too (`aBlock_recip`).

Remark 6 (geometric meaning). The disk model and its exterior $\{|z| > 1\} \cup \{\infty\}$, each with the metric $4|dz|^2/(1-|z|^2)^2$, are two copies of $\mathbb{H}^2$, exchanged by the inversion $\iota(z) = 1/\bar{z}$ in the boundary circle; ι is an isometry between them. It preserves the pair term, since $\left|\frac{1/\bar{z}-1/\bar{w}}{1-1/(\bar{z}w)}\right| = \frac{|z-w|}{|1-z\bar{w}|}$, and it maps the ring of Euclidean radius $\sqrt{\xi}$ to the ring of radius $1/\sqrt{\xi}$. So the continuation of the blocks to $\xi > 1$ describes the same hyperbolic ring in the exterior chart, and $1/\xi_*$ is the threshold seen from that chart, not a second physical threshold. (`derivation.txt` gives only the algebraic proof above. We record this interpretation as a remark and do not use it in any proof. The sign change of g_m reflects the fact that ι reverses orientation.)

Clearing denominators, $A_m := 2(1-\xi)^2(1-X)^2 a_m \in \mathbb{Z}[\xi]$ satisfies $A_m(1/\xi) = \xi^{-2N-2}A_m(\xi)$, so A_m is self-reciprocal. We write $B_m := 2(1-X)^2 b_m$, $G_m := 2(1-X)^2 g_m$ and $D_m := A_m B_m - (1-\xi)^2 G_m^2 = 4(1-\xi)^2(1-X)^4(a_m b_m - g_m^2)$. On $(0,1)$ the signs of a_m, b_m and $a_m b_m - g_m^2$ are those of A_m, B_m and D_m.

Threshold equation. If $b_m > 0$ on $(0,1)$ for all m (certified for $N \le 16$, §7), instability occurs only through the radial entries. The spectral threshold $\xi_*(N)$ is then the largest root in $(0,1)$ of $\prod_m a_m$. For $N \le 16$ this is the unique root of $a_{\lfloor N/2 \rfloor}$ in $(0,1)$. For even N and $m = N/2$, the bracket and the N^2X term combine. With

$$y := \xi^{N/2} = \tanh^N(r/2),$$

the threshold equation $a_{N/2} = 0$ reads

$$(N-1)c - \frac{(c-1)^2}{2} + \frac{2Ncy^2}{1-y^2} = \frac{N^2}{8} + \frac{N^2 y}{2(1-y)^2}, \qquad c = \frac{1+\xi}{1-\xi}. \tag{3}$$

(Use $m\xi^m + (N-m)\xi^{N-m} = Ny$, $(N-m)\xi^{N+m} + m\xi^{2N-m} = Ny^3$, and $y(1+y^2)+2y^2 = y(1+y)^2$.) For odd N we use the general a_m with $m = (N-1)/2$. In the sinh model the right-hand side is the same Thomson term $N^2/8$ without the image correction, and the left-hand side is $(N-1)(1+c^2)/2$.

Algebraicity is automatic; the structure is not. Since the blocks are rational in ξ with rational coefficients, every threshold is a root of an integer polynomial, so "the thresholds are algebraic" carries no information by itself. What is informative is the structure: the self-reciprocal numerators A_m, the irreducible palindromic factor that carries ξ_*, its degree, and its leading and constant coefficients (§7).

The natural invariant. By reciprocity, if the minimal polynomial f of ξ_* is palindromic of degree n, then $f(x) = x^{n/2} g(x + 1/x)$, and $C := \xi_* + \xi_*^{-1}$ has degree $n/2$ with minimal polynomial g. Since $\xi = \tanh^2(r/2)$,

$$C = \xi_* + \xi_*^{-1} = 2 + \frac{4}{\sinh^2 r_*} = \frac{2(\cosh^2 r_* + 1)}{\cosh^2 r_* - 1}.$$

So $\cosh^2 r_*$ generates the same field as C. In the sinh model, $C = 2 + 4/(q-1)$ is rational (`note.tex`, §4).

7 Certified thresholds for $N \le 16$

Method (`certify_v2.py` → `certify_v2.out`). For each $N \le 16$ and $1 \le m \le \lfloor N/2 \rfloor$:

1. Form the exact integer polynomials A_m, B_m, D_m (sympy, from Thm. 2).
2. Isolate all real roots in $(0,1)$ of the square-free parts with exact rational real-root isolation, after removing the factors ξ and $\xi - 1$. This shows: B_m has no root and $B_m(\frac{1}{2}) > 0$; every A_m with $m \ne \lfloor N/2 \rfloor$ has all its roots strictly below the isolating interval of $A_{\lfloor N/2 \rfloor}$; and $A_{\lfloor N/2 \rfloor}$ has exactly one root in $(0,1)$.
3. Factor A_{crit} over $\mathbb{Q}$ (sympy) and select the unique irreducible factor f with a root in $(0,1)$. Its irreducibility is certified separately with PARI/GP (§7.1). Refine an exact rational isolating interval $[\alpha, \beta]$ for its root, of width $< 10^{-45}$.
4. Assert exactly that $A_{\mathrm{crit}}(\alpha) < 0 < A_{\mathrm{crit}}(\beta)$. With step 2 this proves $a_{\mathrm{crit}} < 0$ on $(0, \xi_*)$ and $a_m > 0$ for all m on $(\xi_*, 1)$.
5. Compute Arb balls for the roots of f (python-flint 0.9, 400 bits). Select the real root by exact tests only: the real roots returned by `arb_fmpz_poly_complex_roots` have imaginary part exactly 0. Assert that the whole ball $[\mathrm{mid} - \mathrm{rad}, \mathrm{mid} + \mathrm{rad}]$, converted exactly to rationals, lies inside $[\alpha, \beta]$.
6. For the energy–momentum threshold, take the largest root of $\prod_m D_m$ in $(0,1)$. Isolate it exactly, assert that it is separated from the roots of the other D_m, and compare it exactly with $[\alpha, \beta]$.

No floating-point comparison enters any pass/fail decision. The first version (`certify.py`) selected the Arb root by a float comparison of its midpoint. The second reader flagged this, and version 2 replaces it. All values and polynomials in v1 and v2 agree.

Theorem 7 (CERTIFIED, $2 \le N \le 16$)**.** *(a) For $N \le 7$, a_m, b_m and $a_m b_m - g_m^2$ are positive on all of $(0,1)$ for every m. The ring is spectrally stable and has positive-definite non-rotational Hessian at every radius.*

(b) For $8 \le N \le 16$, $b_m > 0$ on $(0,1)$ for every m. The ring is linearly unstable for $0 < \xi < \xi_*(N)$ and spectrally stable for $\xi_*(N) < \xi < 1$. The instability at threshold is in the mode $m = \lfloor N/2 \rfloor$ (and $N - m$), and every other a_m changes sign strictly below ξ_*.

(c) (irreducibility certified with PARI/GP, §7.1) $\xi_*(N)$ is the root in $(0,1)$ of an irreducible palindromic primitive integer polynomial f_N of degree

$$\deg \xi_*(N) = \begin{cases} 2N-2, & N \text{ odd}, \\ \frac{3}{2}N - 2, & N \equiv 0 \pmod 4, \\ \frac{3}{2}N - 3, & N \equiv 2 \pmod 4, \end{cases} \tag{4}$$

with the values in Table 1.

N	ξ_*	$\sqrt{\xi_*}$	r_*	$\cosh r_*$	deg	lead
8	0.06735622300408126226843 74…	0.2595307747	0.5312104560	1.1444414784	10	1
9	0.11287106323660274043969 40…	0.3359628897	0.6990695396	1.2544637167	16	4
10	0.16611424152623911808175 86…	0.4075711490	0.8653901256	1.3984100696	12	7
11	0.20460117688266929063553 58…	0.4523286160	0.9752480466	1.5144618547	20	10
12	0.24741777680406770917396 50…	0.4974110743	1.0917203545	1.6575169309	16	7
13	0.27986138864390668702999 65…	0.5290192706	1.1775646302	1.7772431147	24	18
14	0.31493896311550662339350 54…	0.5611942294	1.2691494599	1.9194478920	18	23
15	0.34245858351217025660102 26…	0.5851996100	1.3406681986	2.0416335000	28	28
16	0.37168181390604545663672 87…	0.6096571282	1.4167509599	2.1831006077	22	17

Table 1: Certified spectral thresholds of the tanh model. $\xi_* = \tanh^2(r_*/2)$ is the squared Euclidean radius in the Poincaré disk, r_* is the hyperbolic radius, "deg" is the degree of the minimal polynomial of ξ_* and "lead" is its leading coefficient (equal to its constant term, by palindromy); both are certified with PARI/GP (§7.1). The Arb radii are $< 3 \cdot 10^{-153}$. The full minimal polynomials, those of $C = \xi_* + 1/\xi_*$, and the discriminants are in `certify_v2.out` and `irreducibility/irreducibility.out`.

Independent confirmation. The second reader recomputed $\xi_*(8), \xi_*(9), \xi_*(10), \xi_*(12)$ from full $2N$-dimensional Cartesian spectra and found agreement to ≥ 43 digits. A grid scan of the full Jacobian for $N = 3..7$ found $\max |\mathrm{Re}\, \sigma| \le 4 \cdot 10^{-19}$ (NUMERICAL).

Structure of the numerator. For even $N \in \{8, \dots, 16\}$, $A_{N/2}$ factors over $\mathbb{Q}$ as a constant times $(\xi - 1)^4 (\xi^{N/2} + 1)$, times one further factor $(\xi + 1)$ when $N \equiv 2 \pmod 4$, times the minimal polynomial of ξ_*. So $(\xi + 1)^2$ divides $A_{N/2}$ exactly when $N \equiv 2 \pmod 4$, with one copy of $(\xi + 1)$ coming from $\xi^{N/2} + 1$. (An earlier draft counted this copy twice; the corrected form is checked in Lean, `A_crit_factor_N8–N16`.) These cyclotomic factors have roots on $|\xi| = 1$ and are spurious.

7.1 Irreducibility (CERTIFIED by PARI/GP)

Scripts and output are in `tanh/irreducibility/`. The polynomials f_N ($8 \le N \le 16$) are read from `lean/Havelock/TanhAlgebra.lean`; `extract.py` recomputes them from `exact.py` (factor $A_{\lfloor N/2 \rfloor}$ and take the factor with a root in $(0,1)$) and finds the two sources identical. With PARI/GP 2.15.4 (`irreducibility.gp`) we check, for every N:

1. `polisirreducible`$(f_N) = 1$, content 1, f_N palindromic, and degree and leading coefficient as in Table 1 and (4).

2. A second, independent witness. For the first six primes $p \nmid \operatorname{lead}(f_N)\operatorname{disc}(f_N)$ we record the degrees of the irreducible factors of $f_N \bmod p$. A factorisation over $\mathbb{Q}$ into factors of degrees d and $n-d$ would give a sub-multiset of each pattern summing to d. For every N no $0 < d < n$ is attained by all six patterns, so f_N is irreducible. For example, for $N = 8$ the patterns mod $3, 5, 7$ are $(5,5)$, $(2,8)$, (10).

3. The trace polynomial g_N with $f_N(x) = x^{n/2} g_N(x + 1/x)$ is irreducible and agrees with `certify_v2.out`; the discriminants agree too.

Since f_N is irreducible and primitive, it is (up to sign) the minimal polynomial of $\xi_*(N)$ over $\mathbb{Z}$. If $\operatorname{lead}(f_N) \neq \pm 1$, then by Gauss's lemma the monic minimal polynomial $f_N/\operatorname{lead}(f_N)$ is not in $\mathbb{Z}[x]$, so $\xi_*(N)$ is not an algebraic integer.

Mod-p certificates. If f_N is irreducible mod a prime $p \nmid \operatorname{lead}(f_N)$, then f_N is irreducible over $\mathbb{Q}$. Such a prime exists for every N here. The smallest ones are $p = 7, 13, 107, 17, 11, 101, 3, 29, 163$ for $N = 8, \dots, 16$. A certificate of this kind could be checked in Lean; we have not done so.

7.2 $N = 8$: an algebraic unit

Corollary 8 (CERTIFIED; irreducibility by PARI/GP)**.** *$\xi_*(8)$ is an algebraic unit of degree 10 with minimal polynomial*

$$x^{10} - 12x^9 - 37x^8 - 72x^7 - 84x^6 - 88x^5 - 84x^4 - 72x^3 - 37x^2 - 12x + 1.$$

$C = \xi_* + \xi_*^{-1} = 2 + 4/\sinh^2 r_*$ has minimal polynomial $C^5 - 12C^4 - 42C^3 - 24C^2 + 32C + 32$. For $9 \le N \le 16$ the leading coefficients of the (irreducible, primitive) f_N are $4, 7, 10, 7, 18, 23, 28, 17$, so $\xi_*(N)$ is not an algebraic integer.

The polynomial is irreducible (§7.1), monic, with constant term 1, so $\xi_*(8)$ is a unit. The second reader recovered the same polynomial independently by PSLQ from a 110-digit value and confirmed its irreducibility (`review/minpoly8.py`). Numerically, its only real roots are $\xi_*(8)$ and $1/\xi_*(8) \approx 14.846$. Four conjugates lie on $|x| = 1$ and four do not, so $\xi_*(8)$ is neither Pisot nor Salem. In the sinh model $\xi_*(8) = 8 - 3\sqrt{7}$ is also a unit, of degree 2. We do not know whether the coincidence has a reason.

Conjecture 9. *For every $N \ge 8$: $b_m > 0$ on $(0,1)$; the critical mode is $\lfloor N/2 \rfloor$; the factor of $A_{\lfloor N/2 \rfloor}$ carrying ξ_* is irreducible and palindromic, so that $1/\xi_*$ is a Galois conjugate of ξ_*; and its degree is given by (4).*

Reciprocity alone shows that $1/\xi_*$ is a root of $A_{\lfloor N/2 \rfloor}$. It does not show that $1/\xi_*$ is a conjugate of ξ_*. That needs irreducibility of the factor, which is why palindromy of the minimal polynomial is certified only for $N \le 16$. The critical mode and $b_m > 0$ near the threshold were checked numerically up to $N = 800$.

8 Odd N: the Krein window

For even N, $g_{N/2} = 0$, so $D_{N/2} = A_{N/2}B_{N/2}$ and the two thresholds coincide. This is certified: the same irreducible factor carries both. For odd N, $g_{\rm crit} \neq 0$. At ξ_* we have $a_{\rm crit}b_{\rm crit} - g_{\rm crit}^2 = -g_{\rm crit}^2 < 0$, and the Hessian becomes definite only at a larger value $\xi_{\rm EM}$.

Proposition 10 (CERTIFIED, odd $9 \le N \le 15$)**.** *$\xi_{\rm EM}(N) > \xi_*(N)$ strictly (disjoint exact rational intervals). The gaps are $3.220 \cdot 10^{-8}$ ($N = 9$), $1.244 \cdot 10^{-7}$ (11), $1.777 \cdot 10^{-7}$ (13) and $1.886 \cdot 10^{-7}$ (15). On $(\xi_*, \xi_{\rm EM})$ the ring is linearly (spectrally) stable, with distinct purely imaginary eigenvalues, but the critical block $M_{\rm crit}$ has one positive and one negative eigenvalue. Its nonlinear stability there is undecided.*

In the window the two frequencies of the critical block carry opposite Krein signatures. The second reader confirmed this on the coordinate-free Cartesian Hessian (count of negative eigenvalues, Krein signs of the eigenvectors; `review/krein_em.py`), so it is not a coordinate artefact. The energy–momentum method gives no conclusion there: the ring is linearly (spectrally) stable, and nonlinear stability is **undecided**. In the sinh model $g_m \equiv 0$, and no such window exists.

9 Large N (NUMERICAL)

Set $c = \kappa N$. Then $\xi = (c-1)/(c+1)$ and $y = \xi^{N/2} \to u := e^{-1/\kappa}$. Dividing (3) by N^2 and formally letting $N \to \infty$ gives the limit equation

$$\kappa - \frac{\kappa^2}{2} + \frac{2\kappa u^2}{1-u^2} - \frac{1}{8} - \frac{u}{2(1-u)^2} = 0, \qquad u = e^{-1/\kappa}. \tag{5}$$

We *define* κ_∞ as the root of (5) near 0.134:

$$\kappa_\infty = 0.1343122142726281390560\ldots$$

(evaluated with `mpmath.findroot` at 40 digits, `asymptotics.py`; re-solved independently in `review/kappa_check.py`). Status: this is a numerically evaluated constant with an explicit defining equation. It is not a fit. We know no closed form; (5) is transcendental in κ through $u = e^{-1/\kappa}$. Dropping the exponentially small u-terms leaves $\kappa^2 - 2\kappa + \frac{1}{4} = 0$, whose root $1 - \sqrt{3}/2 \approx 0.13397$ is within $3.4 \cdot 10^{-4}$ of κ_∞. What is *not* proved is the asymptotic statement $\cosh r_*/N \to \kappa_\infty$: equation (5) comes from a formal limit of the even-N equation, and neither the uniformity of that limit, nor the persistence of the critical mode $\lfloor N/2 \rfloor$, nor the odd-N case is proved. It is supported numerically for $N \le 800$ (`asymptotics.out`). The correction term below is an empirical fit to the same data:

$$\cosh r_* = \kappa_\infty N + \frac{0.541}{N} + O(N^{-2}) \ \ \text{(fit)}, \qquad r_* = \ln(2\kappa_\infty N) + o(1), \qquad 1 - \xi_* \sim \frac{2}{\kappa_\infty N} \approx \frac{14.89}{N}.$$

The second reader re-solved (5) and checked $\cosh r_*/N$ up to $N = 1024$ (`review/kappa_check.py`).

10 Comparison with the sinh model and the flat disk

	sinh model [1]	tanh model (this note)
pair term	$-\ln\sinh(d/2)$	$-\ln\tanh(d/2)$
radial stiffness	$(N-1)(1+c^2)/2$	$(N-1)c-\frac{1}{2}(c-1)^2+O(\xi^N)$
$s\theta$ coupling	none	$g_m=\frac{N}{2}(W_m-W_{N-m})$
threshold	$\cosh^2 r_*=q(N)\in\mathbb{Q}$	algebraic, degree (4)
$N\le 7$	stable for all r	stable for all r (cert.)
$\xi_*(8)$	$8-3\sqrt{7}$ (unit, deg 2)	unit, deg 10
large N	$\cosh r_*\sim\sqrt{N}/2$	$\cosh r_*\sim\kappa_\infty N$ (num.)
odd-N Krein window	none	width $\sim 10^{-7}$

For $8\le N\le 16$ the tanh threshold is larger:

N	8	9	10	11	12	13	14	15	16
$r_*^{\sinh}$	0.5119	0.6585	0.7954	0.8814	0.9692	1.0317	1.0967	1.1462	1.1978
$r_*^{\tanh}$	0.5312	0.6991	0.8654	0.9752	1.0917	1.1776	1.2691	1.3407	1.4168

Both radii grow logarithmically, so $r_*^{\tanh}/r_*^{\sinh}\to 2$, but slowly. This is consistent with the data to $N=800$ (NUMERICAL).

Interpretation. The models differ by the pair term $\ln\cosh(d/2)$, whose Laplacian is constant. In the sinh model this background gives a radial stiffness quadratic in $\cosh r$, which beats the Thomson destabilisation $N^2/8$ once $\cosh r\sim\sqrt{N}/2$. In the tanh model the interaction is screened by curvature: it decays like e^{-d} beyond one curvature radius. The stiffness is then only linear in $\cosh r$, up to exponentially small image terms, so stabilisation needs $\cosh r\sim 0.134N$.

Flat hard-wall disk (context only). This model has the Euclidean area form, image vortices, and the non-constant Robin self-image term; its ring stability goes back to Havelock [7], which we have not read. It is not treated here. The point is only that the same image Green's function, with a different area form and a non-constant Robin function, is a different stability problem. Neither hyperbolic model describes a flat-bottomed hard-wall condensate.

11 Limitations

- **Literature.** Boatto [2] computed the stability ranges of this model numerically, with the same block structure (§1). Her equations reproduce our values exactly; her plotted $K<0$ curves appear to correspond to a different pair law (§1), a reading a specialist should confirm. Kimura [5] and Hwang–Kim 2013 [6] could not be obtained beyond their abstracts. A human with journal access needs to check (i) which Hamiltonian Kimura uses (the tanh attribution here rests on [1, 3, 4, 2]) and (ii) whether Hwang–Kim 2013 already contains ring stability thresholds for this model. Until then, the novelty claim in §1 is provisional.
- **No human specialist review.** The derivation has been re-derived and checked by an independent AI reader (verdict: confirmed with scope fixes, all applied here). No human expert in point-vortex dynamics has reviewed it.

- **Heuristic limit.** κ_∞ is the numerically evaluated root of (5), which comes from a formal limit; $\cosh r_*/N \to \kappa_\infty$ is not proved, and the $1/N$ correction is a fit.
- **Finite range.** Palindromy, irreducibility, the degree pattern, $b_m > 0$ and the critical mode are certified only for $N \leq 16$ (Conjecture 9). Irreducibility is certified by PARI/GP, not by Lean.
- **Nonlinear stability.** This is stated only via Hessian definiteness and modulo $SO(2,1)$ (Remark 4). In the odd-N Krein window the ring is linearly (spectrally) stable and nonlinear stability is undecided; at the threshold itself it is open.

Files

`tanh/`: `derivation.txt` (full derivation), `blocks.py`, `exact.py`, `check_jacobian.py/.out`, `identities.py/.out`, `certify_v2.py/.out` (certification), `asymptotics.py/.out`, `irreducibility/` (PARI/GP irreducibility certificates: `extract.py/.out`, `irreducibility.gp/.out`, `polys.gp`), and `review/` (second reader). Literature record: `literature.md`.

Acknowledgements and disclosure of AI assistance

This work was developed with substantial assistance from an AI system (Claude, Anthropic). The AI assistance covered the derivations, the computer-algebra, certification and Lean code, the independent re-derivations used as "readers", and the drafting of this note. The author directed the work and takes responsibility for the content. No human expert has yet checked it, and we welcome corrections. We thank Stefanella Boatto, whose paper posed the question for this model, and James Montaldi and Tadashi Tokieda.

References

[1] J. Montaldi, T. Tokieda, Deformation of geometry and bifurcations of vortex rings, in *Recent Trends in Dynamical Systems*, Springer Proc. Math. Stat. 35 (2013) 335–370; arXiv:1210.5662 (quotations and page numbers from arXiv v1).

[2] S. Boatto, Curvature perturbations and stability of a ring of vortices, Discrete Contin. Dyn. Syst. Ser. B 10 (2008) 349–375.

[3] J. Montaldi, C. Nava-Gaxiola, Point vortices on the hyperbolic plane, J. Math. Phys. 55 (2014) 102702; arXiv:1403.2138.

[4] C. Nava-Gaxiola, Point vortices on the hyperboloid, PhD thesis, University of Manchester (2013), `eprints.maths.manchester.ac.uk/2022`.

[5] Y. Kimura, Vortex motion on surfaces with constant curvature, Proc. R. Soc. Lond. A 455 (1999) 245–259 (abstract only; cited via [1, 3, 4, 2]).

[6] S. Hwang, S.-C. Kim, Relative equilibria of point vortices on the hyperbolic sphere, J. Math. Phys. 54 (2013) 063101 (abstract only).

[7] T. H. Havelock, The stability of motion of rectilinear vortices in ring formation, Phil. Mag. (7) 11 (1931) (not read; cited for context only).